SCHOOL OF ENGINEERING

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

B. TECH DEGREE 2nd SEMESTER FIRST INTERNAL EXAMINATIONS, APRIL 2024

23-200-0201B: LINEAR ALGEBRA & TRANSFORM TECHNIQUES

(2023 Scheme)

BRANCHES- CS A & IT A

TIME: 2hrs **MAX. MARKS: 30**

Course Outcomes

On completion of this course the student will be able to:

CO1	Solve linear system of equations and to determine Eigen values and Eigen vectors
CO2	Exemplify the concept of vector space and subspace

PARTA (Answer all questions) **(5 x 2= 10 Marks)**

	Questions	Marks	BL	CO	PO
I (a)	Find the rank of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 5 & 10 & 15 \end{bmatrix}$	2	1	1	1
(b)	Find the characteristic equation of $A = \begin{bmatrix} 10 & 3 \\ 4 & 6 \end{bmatrix}$	2	1	1	1
(c)	Show that (R ,+, .) is a field where R is the set of all real numbers.	2	2	1	1
(d)	Show that {(1,0), (0,1) } forms a basis for the vector space R^2 over R	2	2	2	1
(e)	Is the set of all real numbers R over the set of all complex numbers C, a vector space? Justify your answer.	2	1	2	1

PART B (Answer the following questions choosing either the first OR the second option from each module)

MODULE 1 **(1 x 10= 10 Marks)**

Q.No.	Questions	Marks	BL	CO	PO
II (a)	Solve the system of equations $2x + 3y + 4z = 31, x - 2y + 4z = 4, 3x + y - 3z = 2, 6x + 2y - 6z = 4.$	5	3	1	1
(b)	Find the eigen values and eigen vectors of $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$	5	4	1	1
	OR				
III (a)	Find a and b if the rank of the matrix $A = \begin{pmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & a & b \\ 1 & 1 & -2 & a \end{pmatrix}$ is 2.	5	5	1	1
(b)	Using Cayley Hamilton theorem find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix}$	5	4	1	1
	MODULE 2 **(1 x 10= 10 Marks)**				
IV (a)	Determine whether the vectors (1,3,2),(1,-7,8) and (2,11,-1) are linearly independent.	5	3	2	1
(b)	Let V be the vector space of set of all 2x2 matrices. Prove that dim(V) = 4	5	5	2	1

	OR				
V (a)	Find an orthonormal basis for the subspace W of the vector space R^3 over the field R generated by the vectors (2,1,1) and (1,3,-1).	5	6	2	1
(b)	Let V and W be two vector spaces over the same field F and let T be a linear transformation from V to W, Prove that Ker T is a subspace of V and Im T is a subspace of W.	5	4	2	1

Bloom's Taxonomy Levels

L1 - %	L2- %	L3-%	L4-%	L5-%	L6-%
23	15.4	15.4	23	15. 4	7.5

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